

A. (30pts) We have a liquid feed that is 40.0 mol% methane (C1), 20.0 mol% n-butane (C4), 15.0 mol% n-pentane (C5) and unknown mol% of n-hexane (C6). Feed rate (F) is 100 kmol/h and fraction feed vaporized (V/F) is 0.7. Feed pressure is 100.0 psia. The flash drum operates at 250 kPa and 10 °C.

- 1) (5pts) Sketch the process.
- 2) (5 pts) Find mole fraction (z_4) of n-hexane in the feed and flow rate of vapor stream (V) and liquid stream (L).
- 3) (15 pts) Find V/F use Rachford-Rice Equations. $K_{C1} = 50$, $K_{C4} = 0.6$, $K_{C5} = 0.17$, and $K_{C6} = 0.05$. Use (V/F)₁ = 0.45 as 1st guess and find V/F value when $f(V/f)$ converges to less than 0.01.

$$f\left[\frac{V}{F}\right] = \sum_{i=1}^c \frac{(K_i - 1)z_i}{1 + (K_i - 1)\left(\frac{V}{F}\right)} = 0$$

$$\left[\frac{V}{F}\right]_{k+1} = \left[\frac{V}{F}\right]_k - \frac{f_k}{\left[\frac{df_k}{d(V/F)}\right]}$$

$$\frac{df_k}{d(V/F)} = - \sum_{i=1}^c \frac{(K_i - 1)^2 z_i}{\left[1 + (K_i - 1)\left(\frac{V}{F}\right)\right]^2}$$

- 4) (5 pts) Find x_i , y_i , V, and L

$$x_i = \frac{z_i}{1 + (K_i - 1)\frac{V}{F}} \quad i=1, C \quad y_i = K_i x_i = \frac{K_i z_i}{1 + (K_i - 1)\frac{V}{F}} \quad i=1, C$$

- 5) (5 pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. Make your own flowsheet to find T_{drum} , V/F, x_i , y_i , L, and V

B. (70 pts) We are separating acetone and ethanol in a stripping column. The column has two feeds and operates at a pressure of 1.0 atm. Feed 1 is a saturated liquid and is fed into the top of column (no condenser). Flow rate of F1 is 150.0 kmol/hr with 60.0 mol% acetone. Feed 2 is a 10.0 mol% acetone and two-phase mixture with that is 80% vapor. Flow rate of F2 = 50.0 kmol/hr. Assume CMO is valid. Bottoms product is 4.0 mol% acetone and the column has a partial reboiler with bottom up. Boil-up ratio ($\bar{V}/B$) is 1.5. $x_D = 0.10$ $x_B = 0.04$ $q = 0.20$ MB & 0.4 + 0.6 = 1.0 overall E

- 1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables (use V & L for enriching section and $\bar{L}$ & $\bar{V}$ for stripping section)
- 2) (5 pts) There are total three balance envelopes required for this question (enriching & Feed 1 section, stripping section, and Feed2). Draw separately from (a) and write down mass balance for each section.
- 3) (5 pts) Set up overall mass balance and component mass balance
- 4) (10 pts) Find L, $\bar{L}$, B, $\bar{V}$, V, and D

5) (10 pts) The equation for the feed line is:

$$y = \frac{q_i}{z_i} x + \frac{1 - q_i}{z_i}$$

$$q_i = \frac{L_{Fi}}{F_i}$$

Find q_1 , q_2 , coordinates on $y=x$ line, and slopes of two feed lines (F_1 & F_2), and y -intercept of F_2 .

6) (5 pts) Find mole composition of ethanol in distillate stream (y_D)

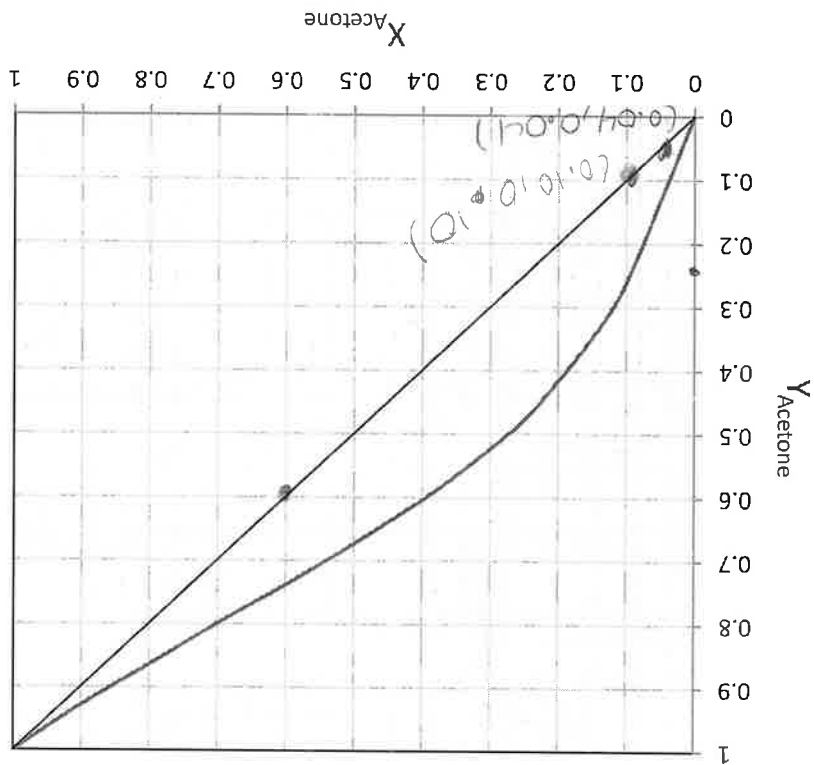
7) (5 pts) Set up component mass balance equation for the top & feed 1 section using mass balance from (b). Then, derive operating equation for the top section.

8) (5 pts) Find the slope, y -intercept, and the intersection point of Feed 1 and top operating line.

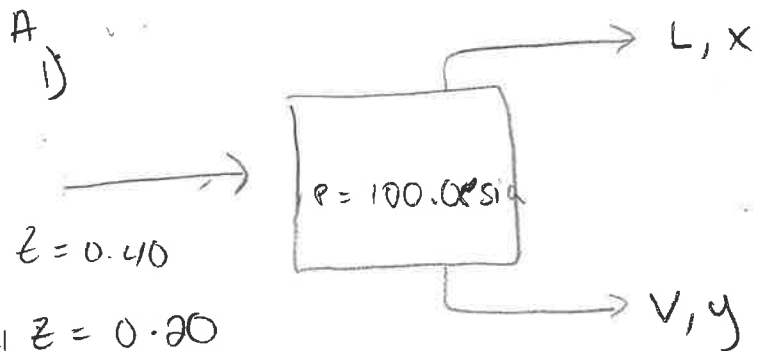
9) (5 pts) Find slope and coordinate on $y=x$ curve of the bottom operating line.

$$y = \left(\frac{L}{V}\right)x - \left(\frac{L}{V} - 1\right)x_B$$

10) (10 pts) Draw the top operating, bottom operating, two feeds, and middle operating lines



11) (5 pts) Find the number of stages and the optimum locations of feed 1 and feed 2 stages. Step off stage from the bottom.



5/5

$$c_1 z = 0.40$$

$$c_4 z = 0.20$$

$$c_5 z = 0.15$$

$$c_6 z = ?$$

$$F = 100 \frac{\text{kmol}}{\text{hr}}$$

$$\frac{V}{F} = 0.7$$

$$\frac{L}{D}$$

$$L' = F \cdot 0.7$$

$$\frac{V}{F} = f$$

2)

$$z_{c1} = 0.4$$

$$z_{c4} = 0.20$$

$$z_{c5} = 0.15$$

$$z_{c6} = 1 - 0.75$$

$$0.75$$

$$z_{c6} = 0.25$$

$$V = 70 \frac{\text{kmol}}{\text{hr}}$$

$$L = 30 \frac{\text{kmol}}{\text{hr}}$$

$$F = L + V$$

$$Fz = Lx + Vy$$

$$y = \frac{L}{V} x + \frac{F}{V} z$$

3)

$$\sum_{i=1}^4 \frac{(K_i - 1) z_i}{1 + (K_i - 1) \frac{V}{F}} = \frac{(50 - 1) 0.4}{1 + (50 - 1) 0.45} + \frac{(0.6 - 1) 0.20}{1 + (0.6 - 1) 0.45} + \frac{(0.17 - 1) 0.15}{1 + (0.17 - 1) 0.45}$$

$$= 0.85 + \frac{-0.08}{0.82} + \frac{-0.124}{0.62} + \frac{(0.05 - 1) 0.25}{1 + (0.05 - 1) 0.45}$$

$$1.807 - 0.047$$

$$0.85 - 0.097 - 0.20 - 0.416$$

$$- \frac{0.1937}{0.57}$$

DO NOTATION CONVERSION

$$= 0.137$$

$$- \sum_{i=1}^4 \frac{(K_i - 1)^2 z_i}{K_i [1 + (K_i - 1) (\frac{V}{F})]}$$

$$- \frac{(50-1)^2 0.4}{[1+(50-1)0.145]} - \frac{(0.6-1)^2 0.20}{[1+(0.6-1)0.137]} - \frac{(0.17-1)^2 0.15}{[1+(0.17-1)0.137]} - \frac{(0.05-1)^2 0.25}{[1+(0.05-1)0.137]}$$

$$- \frac{960.4}{531.30} - \frac{0.032}{0.89} - \frac{0.10}{0.785} - \frac{0.20}{0.75}$$

$$-1.80 - 0.035 - 0.12 - 0.26 \quad 5/15$$

$$4) x_i = \frac{z_i}{1+(k_i-1)\frac{V}{F}}$$

$$y_i = k_i x_i = \frac{k_i z_i}{1+(k_i-1)\frac{V}{F}}$$

$$4) \quad 0/5$$

$$5) \quad 0/5$$

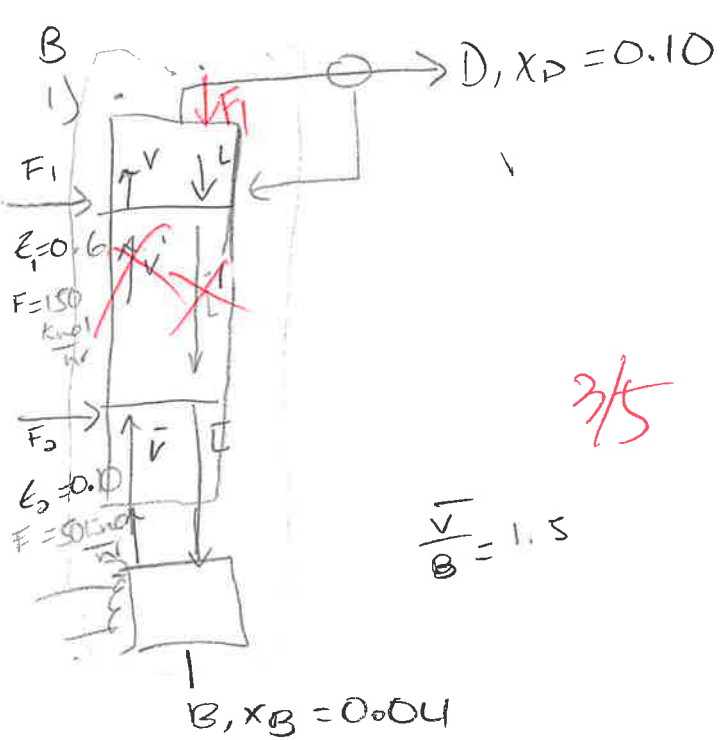
$$\frac{V}{F} \quad 2.16$$

$$0.18 - 0.0915 - 0.157 + 0.225$$

$$\frac{V}{F} \quad 0.15$$

$$\underline{19.6}$$

$$2.39 - 0.085 - 0.14 - 0.27$$

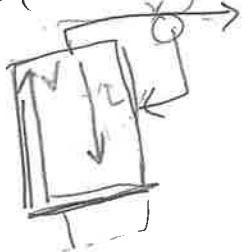


2) Enriching

$$y = \frac{L}{V}x + (1 - \frac{L}{V})x_D$$

$$V = L + D$$

$$F_1 \rightarrow$$



$$(0.10, 0.10)$$

$$I_N = O_{U+}$$

$$F_1 + L = V + D$$

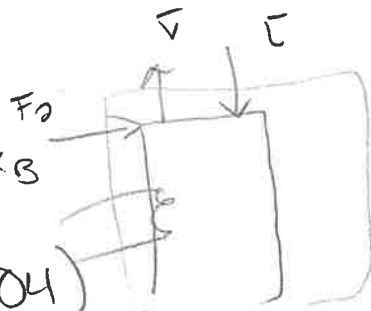
2) 2/5

stripping

$$y = \frac{\bar{L}}{\bar{V}}x + (\frac{\bar{L}}{\bar{V}} - 1)x_B$$

$$\bar{L} = \bar{V} + B$$

$$(0.04, 0.04)$$



$$B, x_B = 0.04$$

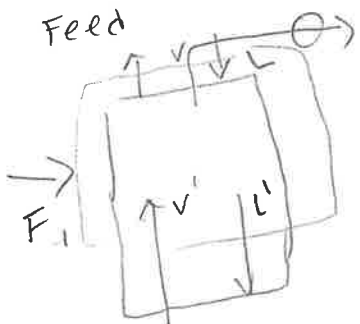
$$m_B = I_N = O_{U+}$$

$$F_2 + \bar{L} = \bar{V} + B$$

$$L = \bar{L}$$

$$V = \bar{V}$$

Feed



$$F_1 + V = L + D$$

$$F_1 z_1 + V y + L x + D x_D$$

$$y =$$

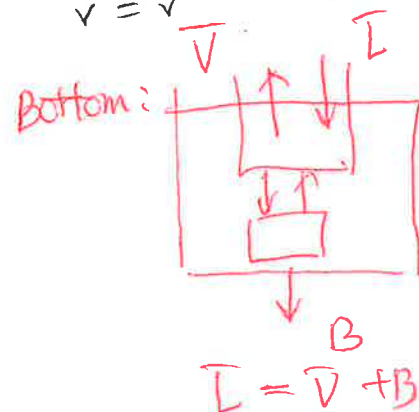
3) overall mB

$$F_1 + F_2 = D + B$$

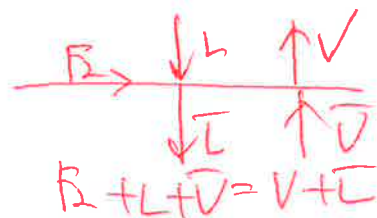
component

$$F_1 z_1 + F_2 z_2 = D x_D + B x_B$$

3) 4/5



Feed 2



4) Feed D. BAIAN &

$$Fz_1 + Vy = Lx + Dx D$$

CMO $\bar{L} = L$

$\bar{V} = V$

STRIPPING

$$y = \frac{\bar{L}}{\bar{V}} x$$

$$\bar{L} = V + B$$

4) 0/10

STRIPPING

$$\text{SLOPE} = \frac{V+B}{V} = \frac{\frac{V}{B} + 1}{V/B} = \frac{1.5 + 1}{1.5} = \frac{2.5}{1.5}$$

$$y = m \frac{\bar{L}}{\bar{V}} - 1$$

$$y = 0.6(x_B)$$

$$\frac{L}{V} \text{ SLOPE} = 1.6$$

$$1.6x + 0.6(0.04)$$

$$y = 1.6x + 0.024$$

STRIPPING

$$5) y = \frac{q}{q-1} x + \frac{z}{1-q}$$

Set $q=1$ ✓

$(0.6, 0.6)$ ✓

Feed 1

slope: ∞

6/10

$$V + L' = T_1 + L + V_1$$

$$LF_2 = L' - L$$

$$q_2 = \frac{LF_2}{F_2}$$

$y=x$ coord for Feed 2
 $(0.10, 0.10)$

$q=0.2$ > slope: $-\frac{1}{4}$

~~0.12~~

$$y = -0.25x + 0.125$$

